

Hydrogen Fuel Assist for Diesel Generators

3–15% fuel savings. Up to 95% PM reduction. No engine modification. Just add water.

THE PROBLEM

India has over 10 million diesel generators powering manufacturing plants, telecom towers, construction sites, mines, ports, commercial buildings, hospitals, and data centres. These DG sets consume billions of litres of diesel annually and are a major source of PM 2.5, CO, HC, NOx, and CO2 emissions. Current CPCB guidelines (up to CPCB IV+) focus primarily on PM and NOx limits for new generators — but do not address CO2 emissions. And millions of legacy DG sets already in operation have no emission retrofit requirement at all.

THE PROBLEM WITH POST-EMISSION RETROFITS

Many industries attempt to address DG emissions using post-treatment systems like Diesel Particulate Filters (DPF), Selective Catalytic Reduction (SCR), and Diesel Oxidation Catalysts (DOC). These aftertreatment systems create significant operational problems:

	Post-Treatment Retrofits	SGT GreenX
Backpressure	Creates exhaust backpressure — engine works harder, burns MORE fuel	Zero backpressure — nothing added to exhaust path
Fuel consumption	Increases fuel consumption by 2–5% due to backpressure	REDUCES fuel consumption by 3–15%
N2O creation	SCR systems generate N2O (nitrous oxide) — a greenhouse gas 265x worse than CO2	Zero N2O — no catalytic chemicals involved
CO2 impact	No CO2 reduction. Higher fuel use = more CO2	Less fuel burned = directly less CO2 emitted
Maintenance	Requires DPF cleaning/replacement, urea/AdBlue refills, catalyst replacement	Water and electrolyte only. Simple, inexpensive.
Engine modification	Exhaust system modification required. Risk of warranty void.	Zero engine modification. Retrofit to air intake only.
Approach	Treats symptoms AFTER combustion	Fixes the root cause: incomplete combustion

THE SGT GREENX SOLUTION

SGT HydroEdge is the first Indian company to manufacture specialized Controlled Hydrogen Fuel Assist technology in their HHO (oxyhydrogen) generators, purpose-built for diesel generators from 25 KVA to 4,000 KVA. GreenX generates hydrogen-oxygen gas on-demand through water electrolysis and injects it into the DG set's air intake in a controlled manner using algorithms. Hydrogen's wide flammability range and extremely low ignition energy act as a combustion catalyst — the diesel burns more completely, less fuel is wasted, and emissions are reduced at source.

GreenX is pre-combustion enhancement, not post-combustion treatment. It fixes the root cause of emissions — incomplete combustion — rather than trying to filter the damage afterwards.

PROVEN RESULTS

3–15%

Fuel Savings

Up to 95%

PM 2.5 Reduction

>50%

CO Reduction

>50%

HC Reduction

Significant

NOx & SOx Reduction

Getting Validated by KOEL (Kirloskar Oil Engines) through active testing partnership. **ARAI (Automotive Research Association of India)** conducting joint testing for Government of India approval as emission reduction technology for all legacy DG sets. Tested across 125 KVA to 1,500 KVA range with positive results backed by stack emission reports from NABL certified labs.

SGT'S PATENT-APPLIED TECHNOLOGY ADVANTAGE

SGT HydroEdge has developed proprietary, patent-applied HHO generator technology that goes far beyond conventional electrolyzers:

High HHO Yield	Patent-applied cell design delivers significantly higher hydrogen-oxygen gas output per unit of electrical input compared to conventional electrolyzers.
Low Power Consumption	Optimised electrode geometry and electrolyte management reduce electricity consumption, maximising net energy benefit.
Long Operational Life	Engineered for continuous industrial duty. Durable materials ensure years of reliable operation without performance degradation for days non stop.
Safety by Design	On-demand generation — zero hydrogen storage. Built-in pressure relief, flashback arrestors, and fail-safe interlock controls
Easy & Low-Cost Maintenance	Modular design for quick servicing. No specialised tools. Consumables: water and standard electrolyte only
Compact Size	Small footprint designed to sit alongside DG sets of any size without requiring additional space or civil works.

DIGITALLY CONNECTED — IOT & DIGITAL TWIN READY

Every GreenX unit is fully digitally connected.

Built-in IoT sensors continuously monitor system performance, HHO output, power consumption, and DG operating parameters. This data feeds directly into GreenVision — SGT's proprietary AI-powered Digital Twin platform.

- **Real-time monitoring** of HHO system health, gas output, and DG fuel efficiency gains
- **Emission tracking** — continuous measurement of CO₂, and lab measurements of PM 2.5, CO, HC, and NOx reductions occasionally
- **ESG-grade reporting** — automated reports for BRSR compliance, investor disclosures, and carbon credit documentation in pipeline
- **Fleet management** — monitor hundreds of DG sets across multiple sites from a single dashboard and control them remotely
- **Predictive maintenance** — AI detects performance degradation for both HHO systems and DG sets before failures occur

WHAT GLOBAL RESEARCH PROVES

Researchers worldwide have extensively studied HHO-enriched combustion in diesel engines. The results are consistent and compelling:

FUEL EFFICIENCY & PERFORMANCE

- Thermal efficiency of diesel engines increases by up to 31.5% with HHO supplementation, with fuel consumption reductions of 12–20% documented across multiple engine sizes and loads
- Brake specific fuel consumption (BSFC) reduced by 12.6–13.7% on average across controlled testing, with some studies showing up to 20% reduction at optimised HHO flow rates
- Engine torque and brake power increased by 9–22.4% depending on engine size and HHO flow rate, demonstrating that HHO does not just save fuel — it improves engine output simultaneously
- A 500-hour continuous operation study confirmed no structural damage to engine components from dual diesel-HHO combustion, validating long-term safety and reliability

EMISSION REDUCTIONS

- Particulate matter and smoke opacity reduced by 25–95% depending on HHO flow rate and engine load — the most dramatic emission improvement of any single intervention
- CO (carbon monoxide) emissions reduced by 19.8–54.5% across studies, with higher reductions at optimised HHO flow rates. One study showed 42.36% CO reduction at 7 LPM HHO flow
- HC (hydrocarbon) emissions reduced by 7.7–46.15% depending on operating conditions, confirming significantly more complete combustion of diesel fuel
- CO₂ emissions reduced by 7–15.2% — a direct consequence of burning less diesel for the same power output. This is the emission metric CPCB does not currently regulate but ESG mandates require
- NO_x behaviour varies by engine and HHO flow rate. Several studies report reductions of 4.3–15.5%, particularly at moderate loads. SGT's testing confirms significant overall NO_x reduction with optimised system calibration

These findings are based on research by scientists and engineers at institutions worldwide including University of Technology Baghdad (Iraq), Wroclaw University (Poland), UNAM Mexico, National University of Sciences and Technology Pakistan, multiple research groups in India, Turkey, Egypt, Ecuador, and Malaysia, published in peer-reviewed journals indexed in ScienceDirect, Springer, MDPI, SAGE, Elsevier, and Nature Scientific Reports. Full references available on request.

APPLICATIONS

Manufacturing

Backup & captive power DG sets

Telecom Towers

Remote DG sets

Construction & Mining

On-site power in harsh environments

Ports & Logistics

Terminal power & yard equipment

Commercial Buildings

Hotels, hospitals, malls, data centres

Campuses & Facilities

University, corporate & govt campuses

DG set range	25 KVA to 4,000 KVA
Installation	Retrofit to air intake. No engine modification.
Hydrogen storage	None — on-demand generation. Zero storage risk.
Input required	Water and electricity only. No chemicals, no urea.
Connectivity	IoT enabled. Integrates with SGT GreenVision Digital Twin.
Technology	Patent-applied. High yield. Low power. Long life.

Ready to Cut Your DG Set Fuel Costs & Emissions?

Ask us for a no-obligation assessment of your DG set operations.

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SGT HydroEdge — Sustainability Infrastructure for Industry